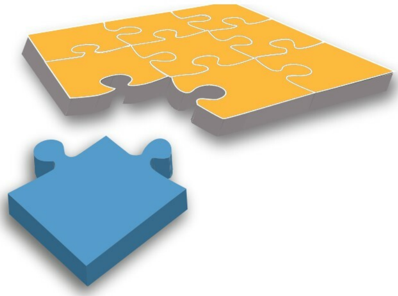

MySQL: Drive Your Performance Problems Away!

Here are some simple steps to tune MySQL for performance and scalability in multi-user environments. Also included are some very effective tools, most of which are packaged along with MySQL.

You've just heaved a sigh of relief after the latest release of your online system—but you can already see troubled times ahead, with performance issues cropping up after adding on just a few concurrent users. You don't have time for elaborate load or scalability testing, since the user count will grow rapidly in the next few weeks. As the solutions architect, you know that you'll get the maximum returns on your investment of effort by taming the database. Will you start scrambling for tools, hoping they will help you? Or will you simply decide to throw more powerful hardware at the system, to postpone solving the problem?

The approach we advocate is to tune your MySQL database system for the most efficient performance before you begin throwing new hardware at the problem, or begin re-designing the very solution itself. Let's learn to use some very effective tools—most of which are included in your MySQL installation—to make this process a pleasure. We can start right away if you are ready to take the plunge.

Find the trouble-makers

Our context here is a production system with real users. Therefore, logging information related to the users' access patterns, and the corresponding system performance, must be available. MySQL has different kinds of logs. One of these is the *slow query* log, which logs details of queries that have exceeded a specified threshold execution time. So here, we will use this slow query log to identify slow-running queries. The steps are as follows:

1. **Decide when to log:** Usually, logs are considered detrimental to system performance, especially in a production scenario. However, since this information is vital for performance optimisation, you could enable logging only for a representative period of time—for example, peak business hours, when the maximum users are logged in, and there is a likelihood of performance problems surfacing.
2. **Enable the log and set the threshold value**: Add the following statements